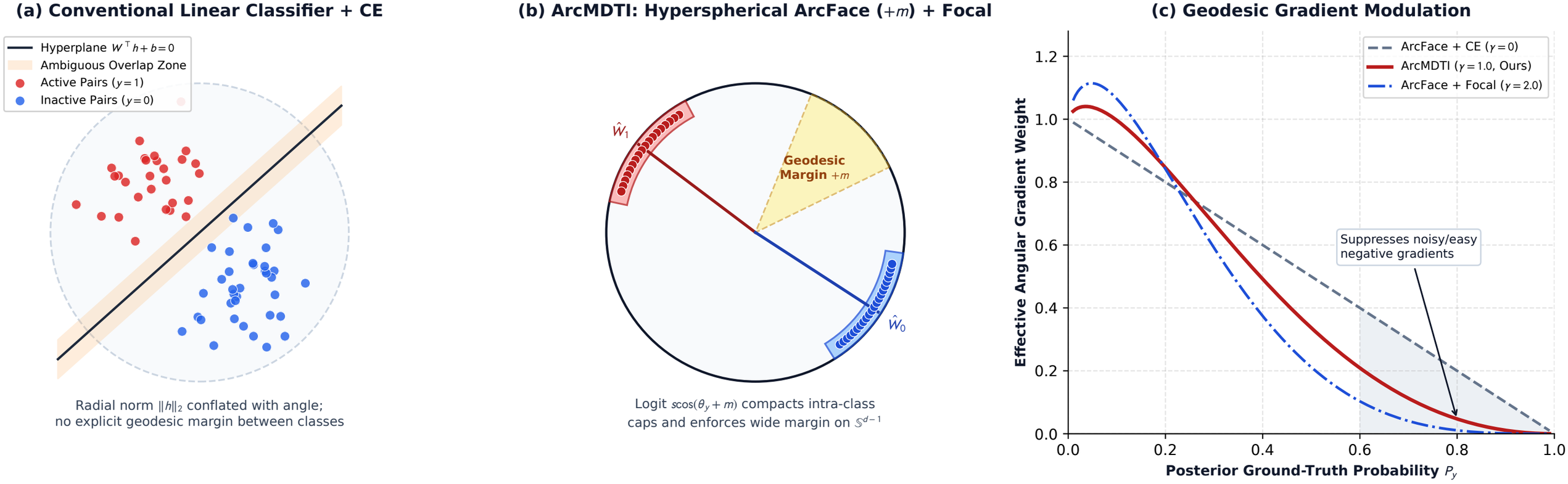
(a) Conventional Linear Classifier + CE
Hyperplane $W^\top h + b = 0$
Ambiguous Overlap Zone
Active Pairs ($y = 1$)
Inactive Pairs ($y = 0$)
Radial norm $\|h\|_2$ conflated with angle;
no explicit geodesic margin between classes
(b) ArcMDTI: Hyperspherical ArcFace ($+m$) + Focal
$\hat{W}_1$
Geodesic
Margin $+m$
$\hat{W}_0$
Logit $s\cos(\theta_y + m)$ compacts intra-class
caps and enforces wide margin on $\mathbb{S}^{d-1}$
(c) Geodesic Gradient Modulation
ArcFace + CE ($\gamma = 0$)
ArcMDTI ($\gamma = 1.0$, Ours)
ArcFace + Focal ($\gamma = 2.0$)
Suppresses noisy/easy
negative gradients
Effective Angular Gradient Weight
Posterior Ground-Truth Probability P_y
0.0
0.2
0.4
0.6
0.8
1.0
1.2